

Signals and Systems Test 3 (7/13/2026)

It will be based on Homework 5, 6, and 7 for chapter 4. You need to know the following topics.

- 1- Given the differential equation of a system, the input $x(t)$, you should be able to find the transfer function $H(s)$ and the zero-state response.
- 2- Given the system differential equation or the transfer function $H(s)$, you should be able to find the frequency response $H(j\omega)$, its magnitude and phase. If a dc or sinusoidal input is given, then you should be able to find the output $y(t)$ and the steady state response.
- 3- Given the system differential equation or the transfer function $H(s)$, you should be able to find the poles and zeros of the system and determine its external and internal stability.
- 4- Given a transfer function $H(s)$, you should be able to draw the canonical Form II realization, cascade and parallel realization, block diagram with amplifier, integrator, and summing amplifier.
- 5- Given the transfer function $H(s)$, you should be able to draw the poles and zeros on the s-plane and draw a rough plot of the magnitude of the frequency response $|H(j\omega)|$.
- 6- Design a filter that meets determined specifications or a given plot of $|H(j\omega)|$.